WebClient Computing

HTML Basics

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Announcements

- Homework-0
  - Create your own branch in kmu-leeky/kmu-cs-web-client-161 repository
  - The due is by the end of Friday
- Another tutorial on the GitHub fork will be made in the next Thursday class
- HTML book recommendation – this is not a textbook
  - (Head first) HTML and CSS : HTML5를 적용한 웹 제작 지침서
  - HTML & CSS : 웹사이트 개발과 디자인 기초
  - HTML5 + CSS3 + 자바스크립트의 정석 : HTML5 필수 API로 배우는 차세대 웹 프로그래밍

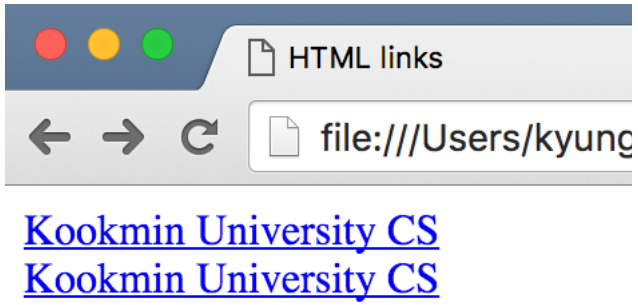
HTML – Make a Link to Other Page

- Hyper Text
  - a structured text that uses links to navigate through different pages
- WWW – World Wide Web
  - Interlinked by hyper text
- Anchor tag <a> represents a link
- Destination is added as an attribute within <a> tag
  - <a href="cs.kookmin.ac.kr">Kookmin Univ. Computer Science</a>

HTML Link Example

- To open the link in a new window add target="_blank" attribute

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML links</title>
</head>
<body>
  <a href="http://cs.kookmin.ac.kr">Kookmin University CS</a><br>
  <a href="http://cs.kookmin.ac.kr" target="_blank">Kookmin University CS</a>
</body>
</html>
```



HTML – Link within a Page

- Within a page, we can assign ID to an entry and link to it

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML page links</title>
</head>
<body>
  <p id="bookmark-1">This is a new paragraph</p>
  <br><br><br><br><br><br><br><br><br><br><br><br>
  <br><br><br><br><br><br><br><br><br><br><br><br>
  <br><br><br><br><br><br><br><br><br><br><br><br>
  <br><br><br><br><br><br><br><br><br><br><br><br>
  <br><br><br><br><br><br><br><br><br><br><br><br>
  <a href="#bookmark-1">jump in a page</a>
</body>
</html>
~
```

Insert an Image

- Using <img> tag. Attributes
  - src : the location of image file. Both local file system path and URL are OK
  - width : the width of the image in a page (unit: pixel or %)
  - height : the height of the image in a page (unit: pixel or %)
  - title : a message shown when a pointer is over the image
  - alt : a message that will be shown when the image is not available
  - <img src="http://www.kookmin.ac.kr/images/common/footer_logo.gif" width=30 height=10 alt="KMU">
- HTML Screen reader
  - A software program that read what is displayed on the screen
  - For blind people, visually impaired, learning disabled

HTML Image Example

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
  <img src="http://www.kookmin.ac.kr/images/common/footer_logo.gif" width="150" height="50" title="KMU Logo"></br>
  <img src="foo.bar.invalid.address" width="150" height="50" alt="KMU Logo"><br>
  <img src="http://www.kookmin.ac.kr/images/common/footer_logo.gif" width="20%" height="auto">
</body>
</html>
~
```

KMU Logo

HTML Image Map

- A image map with clickable areas
- img tag usemap attribute is matched with the map tag name
- map tag contains multiple area tags
  - coords
    - rect: coordinate of rectangle
    - circle: center coordinate and radius

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
  <img src="planets.gif" alt="Planets" usemap="#planetmap">
  <map name="planetmap">
    <area shape="rect" coords="0,0,82,126" alt="Sun" href="basic.html">
    <area shape="circle" coords="90,58,3" alt="Mercury" href="color.html">
    <area shape="circle" coords="124,58,8" alt="Venus" href="fonts.html">
  </map>
</body>
</html>
```

HTML5 Video Tag

- Used to add a video into a web page
- Under the video tag, nest source tag with src specifying the movie location
  - controls – show control bar
  - loop – loop when the file play is over
  - autoplay – automatically start the rendering when the page is ready
  - <source> tag contains type to specify the video type

```
<!DOCTYPE html>
<html>
<head>
  <title>Adding moive to a page</title>
</head>
<body>
  <video width="640" height="480" controls loop>
    <source src="movie.mp4" type="video/mp4">
  </video>
</body>
</html>
```

HTML5 Audio Tag

- Very similar to <video> tag
- Audio type
  - MP3 - audio/mpeg
  - Ogg – audio/ogg
  - Wav - audio/wav

```
<!DOCTYPE html>
<html>
<head>
  <title>Adding moive to a page</title>
</head>
<body>
  <audio controls>
    <source src="horse.mp3" type="audio/mpeg">
  </audio>
</body>
</html>
```

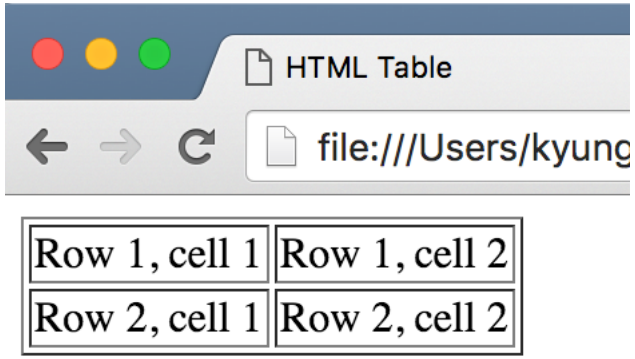
HTML5 Embed Tag

- Adding contents that are not supported with <img><audio><video>
  - The embedded file type has to be defined in the type attribute
    - Flash: appliction/x-shockwave-flash, PDF: application/pdf
    - Application media type list - http://www.iana.org/assignments/media-types/media-types.xhtml#application
  - Be cautios for the copyright.

```
<!DOCTYPE html>
<html>
<head>
  <title>Adding flash to a page</title>
</head>
<body>
  <embed type="application/x-shockwave-flash" src="bigfish.swf" width="600" height=400"></embed><br>
  <a href="http://www.crazymonkeygames.com/Big-Fish.html">The original location</a><br>
  <embed type="application/pdf" src="week1-intro.pdf" width="800" height="1600"</embed>
</body>
</html>
```

HTML Tables

- <table> defines a table
  - Attribute border decides whether the border is drawn
- <tr> defines each row – Table Row
- <td> defines each column with a row – Table Data
  - Data should be written in this field
- Table border attribute
  - Defines border style
  - Styling better to use CSS




```
<!DOCTYPE html>
<html>
<head>
  <title>HTML Table</title>
</head>
<body>
<table border="1">
  <tr>
    <td>Row 1, cell 1</td>
    <td>Row 1, cell 2</td>
  </tr>
  <tr>
    <td>Row 2, cell 1</td>
    <td>Row 2, cell 2</td>
  </tr>
</table>
</body>
</html>
```

HTML Table Heading

- Use Table Heading with <th> tag
  - Bold and centered
- padding attribute – a style attribute to specify the space between table cell and the data within the cell
- Caption tag immediately after the table tag

```html
<!DOCTYPE html>
<html>
<head>
  <title>HTML Table</title>
</head>
<body>
<table border="1">
<caption>Points of employee</caption>
  <tr>
    <th style="padding:15px;">Firstname</th>
    <th style="padding:15px;">Lastname</th>
    <th style="padding:15px;">Points</th>
  </tr>
  <tr>
    <td>Eve</td>
    <td>Jackson</td>
    <td>94</td>
  </tr>
</table>
</body>
</html>
```

Points of employee

Firstname	Lastname	Points
Eve	Jackson	94

Table Cells that Span Many Columns

- colspan tag to make cell to merge multiple columns

Name	Telephone	
Bill Gates	555 77 854	555 77 855

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML Table</title>
</head>
<body>
<table border="1">
  <tr>
    <th>Name</th>
    <th colspan="2">Telephone</th>
  </tr>
  <tr>
    <td>Bill Gates</td>
    <td>555 77 854</td>
    <td>555 77 855</td>
  </tr>
</table>
</body>
</html>
```

Table Cells that Span Many Rows

- rowspan tag to make a cell to merge multiple rows

Name:	Bill Gates
Telephone:	555 77 854
	555 77 855

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML Table</title>
</head>
<body>
<table border="1">
  <tr>
    <th>Name:</th>
    <td>Bill Gates</td>
  </tr>
  <tr>
    <th rowspan="2">Telephone:</th>
    <td>555 77 854</td>
  </tr>
  <tr>
    <td>555 77 855</td>
  </tr>
</table>
</body>
</html>
```

Table Rowspan and Colspan In One Place

two rowspan	four colspan			
	I	II	III	IIII
0	1	2	3	4

```html
<!DOCTYPE html>
<html>
<head>
  <title>HTML Table</title>
</head>
<body>
<table border="1">
  <tr>
    <th rowspan="2">two rowspan</th>
    <th colspan="4">four colspan</th>
  </tr>
  <tr>
    <th>I</th>
    <th>II</th>
    <th>III</th>
    <th>IIII</th>
  </tr>
  <tr>
    <td>0</td>
    <td>1</td>
    <td>2</td>
    <td>3</td>
    <td>4</td>
  </tr>
</table>
</body>
</html>
```

How To Make This?

2 and 2		four colspan			
		first col	second col	third col	fourth col
two rowspan	Third row	1	2	3	4
	Fourth row	5	6	7	8

References

- http://www.w3schools.com/html/ - HTML tutorial
- http://www.htmldog.com/guides/html/
- http://fromyou.tistory.com/581